

01 Sets (Set Concepts)

Mathematics

Big question: How many stones are in my hand?

Think silently → pair talk → share one idea

Today's mission

Goal: I can distinguish between finite and infinite sets usin...

1

You will use: Can We Count Everything?, Defining Our Sets

2

Success: explain, try, and answer in your own words.

3

Write one clear response.

4

Write one clear response.

Starter: think first

Think: How many stones are in my hand?

1

Pair: compare your first answer with a partner.

2

Share: connect your answer to Can We Count Everything?.

3

Write one clear response.

4

Write one clear response.

Can We Count Everything?

Key idea: Concrete exploration of countable objects (stones) vs...

1

Practical counting of stones.

2

Attempting to count grains of sand.

3

Identifying which count can be finished.

4

Write one clear response.

Defining Our Sets

Watch the example.

1

Finite Set: Has a countable number of members.

2

Infinite Set: Has members that go on forever.

3

Examples from daily life.

4

Write one clear response.

Try it together

Can you tell me exactly how many grains are in that pile of sand?

1

Which count was easier to finish?

2

What is a finite set?

3

Agree on one answer before writing.

4

Write one clear response.

Activity challenge

Group task:

1

In groups of five, learners count a pile of 10 bottle caps and th...

2

Students write the definitions of finite and infinite sets in the...

3

Students draw two Venn diagrams. In the first, they represent the...

4

Write one clear response.

Exit ticket

Answer: Can a very large set still be finite? (Yes, if it has an...

1

Use one complete sentence.

2

Include one key word: Set, elements, counting

3

Hand in or read your answer aloud.

4

Write one clear response.